

What are the different MAC protocols available

Bachelor of Technology
Computer Science and Engineering

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1. Introduction

MAC protocols are responsible for controlling how multiple devices share a common communication medium (like a shared cable or wireless channel). They decide who can transmit and when, reducing collisions and improving efficiency.

MAC protocols are mainly classified into three major categories:

- Random Access Protocols
- Controlled Access (Reserved Access) Protocols
- Channelization Protocols

2. Categories

2.1. Random Access Protocols - In these protocols, devices transmit data whenever they have data to send. If a collision occurs, they retransmit after some delay.

Types:

- ALOHA
 - Simplest protocol
 - Devices send data anytime
 - High collision probability
- Slotted ALOHA
 - Time divided into slots
 - Transmission only at slot beginning
 - Better efficiency than pure ALOHA
- CSMA (Carrier Sense Multiple Access)
 - Device checks channel before sending
 - Reduces collisions
- CSMA/CD (Collision Detection)
 - Used in traditional Ethernet
 - Detects collision and stops transmission
 - Example: Wired LAN

- CSMA/CA (Collision Avoidance)
 - Used in wireless networks
 - Avoids collision using acknowledgments
 - Example: Wi-Fi

2.2. Controlled Access (Reserved Access) Protocols - In these protocols, devices take turns to transmit, reducing collisions.

Types:

- Token Passing
 - A special token circulates
 - Only token holder can transmit
 - Used in Token Ring

- Polling
 - Central controller asks devices one by one
 - No collision
 - Used in master-slave systems

- Reservation
 - Devices reserve time slots before transmission
 - Suitable for heavy traffic

2.3. Channelization Protocols - In these protocols, the channel is divided into smaller parts so multiple users can transmit simultaneously.

Types:

- FDMA (Frequency Division Multiple Access)
 - Channel divided into frequency bands
 - Each user gets separate frequency

- TDMA (Time Division Multiple Access)

- Channel divided into time slots
 - Each user gets fixed time
- CDMA (Code Division Multiple Access)
- All users share same channel
 - Different unique codes used
- OFDMA
- Advanced version of FDMA
 - Used in 4G/5G networks

3. Comparison table

Feature	Random Access Protocols	Controlled Access Protocols	Channelization Protocols
Basic Idea	Nodes transmit whenever they have data	Nodes take turns to transmit	Channel divided into parts (time/frequency/code)
Collision	Possible	No collision	No collision
Efficiency (Low Traffic)	High	Medium	Low (wasted slots/bandwidth)
Efficiency (High Traffic)	Low (many collisions)	High	High
Delay	Unpredictable	Predictable	Fixed / Scheduled
Complexity	Simple	Moderate	High
Fairness	Not guaranteed	Guaranteed	Guaranteed
Synchronization Needed	No	Sometimes	Yes (especially TDMA)
Best Used For	Light or burst traffic	Heavy and organized traffic	Continuous high data communication
Examples	ALOHA, CSMA/CD, CSMA/CA	Token Passing, Polling	FDMA, TDMA, CDMA

4. Conclusion

Random access protocols are simple and suitable for light traffic but may suffer from collisions. Controlled access protocols avoid collisions by allowing orderly transmission. Channelization protocols divide the channel into multiple parts, ensuring no collision but requiring higher complexity and synchronization. The choice depends on traffic load and network design requirements.

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